

**IMAGE PROCESSING & COMPUTER VISION (IPCV)
IN-COURSE ASSESSMENT (INDIVIDUAL ASSIGNMENT)
PART 2 – ASSIGNMENT**



Intake : APUMF2604AI
Lecturer : Assoc. Prof. Dr. V. Sivakumar
Due date : 9th August, 2026

1.0 INDIVIDUAL ASSIGNMENT: THIS ASSIGNMENT CARRIES 60% OF YOUR TOTAL ASSESSMENT GRADE.

ASSIGNMENT 2: (60%)

University Professional Attire (Dress Code) Identification

Within university environments, maintaining a professional atmosphere is crucial for fostering a conducive learning environment and preparing students for future professional endeavours. One aspect of this professionalism is adhering to a prescribed dress code. However, ensuring compliance with dress code standards can be challenging, especially in large university settings where manual inspection processes are time-consuming and subject to human error and bias.

The current methods of dress code enforcement often rely on visual inspection by staff or security personnel. This process is not only labor-intensive but also prone to inconsistencies in judgment. Moreover, with the diverse range of attire styles and interpretations, ensuring uniform compliance becomes increasingly difficult.

To address these challenges, there is a growing need for an automated system that can accurately identify and assess adherence to university dress code standards. Such a system would not only streamline the enforcement process but also provide an objective and consistent evaluation of attire worn by individuals within the university premises.

The objective of this assignment is to develop a sophisticated system for automatically identifying and assessing adherence to the university's professional dress code standards. By leveraging image processing and computer vision techniques, students will create a robust solution capable of analyzing attire worn by individuals in various university settings, including classrooms, libraries, and common areas.

Appropriate Attire Conditions

The university's professional dress code encompasses several key criteria that define appropriate attire:

- Long or short-sleeved collared shirts, neatly tucked in.
- Bottom wear such as slacks, trousers, or khakis, presents a formal appearance.
- Footwear options include loafers, sneakers, sports shoes, and boots, contributing to a polished look.
- Office wear standards apply to both male and female individuals, ensuring a gender-neutral approach to dress code enforcement.

Inappropriate Attire Conditions

Conversely, attire that does not meet the university's professional dress code standards includes:

- Open-toed footwear such as sandals (for males), slippers, flip-flops, or slip-ons, which detract from a professional appearance.
- Tops or blouses that are revealing or overly casual, including bare-back, off-shoulder, crop-top, deep-V, spaghetti strap, or tank-top styles.
- Bottom wear such as jogging pants, cargo pants, yoga pants, gym tights/leotards, or sports tights, which are deemed too casual for professional settings.
- Attire typically associated with leisure or recreational activities, such as beachwear, is considered inappropriate within the university environment.
- Garments like skorts (shorts with front skirt panels) are not conducive to a professional appearance and thus do not meet dress code standards.

- Excessive body piercings, excluding those limited to ears and nose, are not aligned with the university's professional image.
- Clothing items like ripped or torn jeans are deemed unsuitable for formal occasions and are therefore prohibited.
- Casual tops such as round neck t-shirts are not considered appropriate for professional settings.
- Accessories like caps, hats, or non-customary headgear are discouraged as they detract from a polished appearance.



Figure 1: Sample Segmentation and Identification of Dress Design for University Attire Identification

2.0 OBJECTIVES OF THIS ASSIGNMENT:

Equip students with the key skills required exploration of knowledge about image processing and computer vision in programming and computing. The focus will be on providing both a high-level understanding and analysis of reusable algorithms and coding techniques that apply to the development of applications.

Project specific objectives:

The primary goal of this assignment is to equip students with the knowledge and skills necessary to develop an advanced system for automatically detecting and evaluating adherence to university dress code standards. By utilizing image processing and computer vision techniques, students will create a solution capable of accurately identifying appropriate and inappropriate attire within university settings.

3.0 DESIRED LEARNING OUTCOMES:

At the end of this in-course assessment (Assignment Part 2), the students should be able to:

- ✓ Evaluate image processing techniques for identifying attire.
- ✓ Implement appropriate image processing techniques for analyzing attire.
- ✓ Interpret the developed solution in comparison to contemporary approaches in image processing.

4.0 DELIVERABLES:

In addition to the system development, a supporting document is needed to reflect on the methodologies and implementation tasks applied in the development of the application. In terms of documentation, you are required to describe and justify the theoretical information about the imaging algorithms used. To fulfill the objective of this coursework, you are also required to describe the results of testing using various test images. Pay particular attention in providing critical analysis for cases of images that fail.

- You should submit an electronic copy of the assignment online.

4.1 PROTOTYPE APPLICATION

- ❖ An individual ScreenCam demonstration of the completed application is to be submitted. Presentation should be done individually by every member of the group and the presentation will last approximately 10-15 minutes.
- ❖ All relevant source code is to be provided. The code should be well commented to enable ease of understanding.
- ❖ All test files (e.g. *.jpg, *.gif, etc.) are to be provided.

Document: Assignment 2 – Report

- ❖ Project final documentation which includes the following as minimum requirements:
 - Table of Contents, etc.
 - Abstract
 - Problem Statement (analysis of the problem)
 - Objectives
 - Proposed Algorithm with Implementation
 - Experimental Results
 - Critical Evaluation & Analysis
 - Conclusion and Recommendations
 - References
 - Appendix

- ❖ Total recommended number of words for documentation is approximately:
 - Assignment 2 is 3000-3500 words

- ❖ All references must be cited in-text and listed at the end of the report using the APA Naming Convention.

Note:

- Do not include programming codes in your documentation (however, you could include parts of the important codes in the Appendix if needed).
- You may source information from the Internet. If you have accessed the Internet, cite the resources carefully in your document.

5.0 DEVELOPMENT TOOLS

The program written for this assignment may be written in any preferable programming language either as below:

	Languages / Tools / Platforms
1	MATLAB
2	Python

The writing of code and the work behind it must belong substantially to you. If any person besides you contributes in any significant way to the assignment, you must credit their work in your description. Similarly, if you include information/source code that you have used from other published sources or web pages, you must cite them as references.

6.0 SUBMISSION REQUIREMENTS:

- ✓ Your report must be typed using Microsoft Word with Times New Roman font, size 12 with 1.5 spaces.
- ❖ Every report must have a *front cover*. The front cover should be presented with the following details:
 - a) Student Name and Student ID number
 - b) Intake code
 - c) Module Code and Module Name
 - d) Assignment Title
 - e) Date Completed (the date the report is due to be handed in)
 - f) Lecturer Name

7.0 TIMELINE

You are required to submit the individual assignment 2 report online on or before **11:59 pm, 9th August 2026**